Internship Report: ShadowFox Virtual Internship

Role: Data Analyst | Level: Intermediate | Duration: 1 September to 30 September 2026

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1. Introduction

I joined the ShadowFox Virtual Internship as a Data Analyst. I completed the beginner project (a sales dashboard) and the intermediate project (customer and revenue analysis). This report covers both, with the focus on the intermediate level.

2. Beginner Level Summary

I built an Excel sales dashboard with a cleaned dataset, KPIs, monthly, category and region trends, charts and recommendations. The data was a sample retail dataset for 2025.

3. Intermediate Level: Objectives

- Clean a realistic dataset and prepare it for analysis.
- Study customer behaviour, repeat purchases and revenue concentration.
- Compare segments, categories, regions and months.
- Explain why the patterns may occur and suggest actions.

4. Dataset and Tools

I used a sample retail dataset for 2025 with 2085 order lines, 799 customers, 12 products, 3 categories, 3 customer segments and 4 regions. The data is for practice and is not real company data. I used Microsoft Excel with SUMIFS, COUNTIFS, INDEX-MATCH and charts. The work is saved on GitHub.

5. Data Cleaning

The raw data had 2132 rows. After cleaning, it has 2085 rows. The steps were:

- Removed 20 duplicate rows.
- Trimmed spaces and fixed text case in 26 Region cells.
- Labelled 10 missing Regions as "Unspecified".
- Removed 15 rows with no CustomerID and 12 rows with no Qty.
- Added Revenue and Profit formulas, and a Customers sheet with one row per customer.

6. Key Findings

Area	Result
Revenue / Profit / Margin	INR 205.7 lakh / INR 62.5 lakh / 30.4%
Orders / Customers	2085 / 799
Repeat customers	58.2% of customers, 88.3% of revenue
One-time customers	41.8% of customers, 11.7% of revenue
Concentration	Top 10% give 42.4%, top 20% give 62.3%, bottom 50% give 10.2%
Revenue per customer	Corporate INR 0.87 lakh, Small Business INR 0.25 lakh, Consumer INR 0.14 lakh
Seasonality	Best Dec, weakest Jul, Oct-Dec give 28.7% of revenue
Category margin	Electronics 27.8%, Furniture 28.5%, Stationery 48.7%
Top region	South, 33.3% of revenue

7. Possible Reasons

- Corporate buyers order larger quantities, so a small group lifts revenue per customer.
- November and December are strong months, and Q4 discounts may help volume.
- Many one-time buyers suggest weak follow-up after the first purchase.

8. Recommendations

- Protect the top 10% of customers with account managers and early offers.
- Send a second-purchase offer to one-time customers.
- Grow the Corporate segment with bulk-order pricing.
- Plan stock for November and December. Run an offer in July.
- Bundle Stationery, which has the best margin, with larger orders.

9. Challenges and Learning

The main challenge was missing customer IDs and duplicates, which change customer counts. I learned to log each cleaning step and to link customer-level tables to order data with formulas. I also learned to measure repeat rate and revenue concentration, and to explain results in simple words.

10. Conclusion

The internship gave me practice in the full analysis flow, from raw data to a dashboard and recommendations. I plan to use these skills in my academic projects.

GitHub repository: [paste your repository link here]